



CONTACT

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GITHUB
github.com/ferdousiMahmudMoon2006

PORTFOLIO
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TECHNICAL SKILLS

PROGRAMMING

C++ C Programming Fundamentals

Basic Data Structures File Handling

Problem Solving

HARDWARE & ROBOTICS

Arduino Embedded Systems Robotics

Sensors & Automation

LANGUAGES

Bangla

Native

English

Professional Working Proficiency

FERDOUSI MAHMUD MOON

Computer Science & Engineering Student

Technology & Innovation Enthusiast

ABOUT

Computer Science & Engineering student at Daffodil International University with a strong interest in technology, innovation and practical problem solving. Experienced in academic and hands-on technology projects spanning robotics, embedded systems and software development, including a smart helmet safety system that earned a Gold Medal at the Innovation World Cup National Round. Active in university clubs and extracurricular activities, with experience in technology competitions and innovation projects, and a growing foundation in C++, Arduino and sensor-based automation.

EDUCATION

B.Sc. in Computer Science and Engineering

Daffodil International University

CGPA 3.25 / 4.00

Higher Secondary School Certificate (HSC)

Barishal Govt. Women's College

GPA 5.00 / 5.00

Secondary School Certificate (SSC)

Barishal Govt. Girls High School, Barishal

GPA 5.00 / 5.00

PROJECTS

PROJECT 01

Smart Helmet Safety System

GOLD MEDAL Life Science Category, Innovation World Cup (IWC), National Round

Developed a smart helmet-based safety system designed to enhance motorcycle rider safety. The system integrates alcohol detection and helmet-wearing detection to prevent the motorcycle from starting when the rider is not wearing a helmet or when alcohol is detected. It also includes accident detection and an emergency alert feature that automatically sends an alert to a predefined emergency contact number in the event of an accident.

Safety Automation Sensors Alcohol Detection Helmet Detection Accident Detection

Emergency Alert System Embedded Systems

PROJECT 02

Flood Detection System

Developed an Arduino-free flood detection system using a water sensor, LED, buzzer, battery and switch to detect rising water levels and provide immediate visual and audio alerts.

PROJECT 03

Smart Stick for the Visually Impaired

Developed a smart walking stick using ultrasonic sensors and a buzzer to detect obstacles and provide audio alerts, helping visually impaired individuals navigate more safely and independently.

PROJECTS — CONTINUED

UNIVERSITY PROJECTS

Library Management System C++ Menu-driven library management system for managing books, issuing and returning books, and maintaining book records using file handling and basic data structures.	Cafe Management System C++ Menu-driven cafe management system for managing food items, customer orders, billing and sales records using structured programming and file handling.
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EARLY PROJECTS & EXPERIMENTS

- Smart Automatic Dustbin**
Arduino-based smart dustbin prototype using an ultrasonic sensor to detect nearby objects and automatically open the lid, providing hands-on experience with sensors, automation and Arduino programming.
- Line Following Robot (LFR)**
Arduino-based autonomous robot designed to detect and follow a predefined path using IR sensors, providing practical experience in robotics, sensor integration and basic control programming.
- Robo Soccer**
Small autonomous robotic system designed for a soccer-based competition, integrating motors, sensors and control logic for movement and ball-handling tasks.
- Fire Alarm System**
Early safety-system prototype using a flame/temperature sensor to detect fire or abnormal heat and activate a buzzer as an immediate warning.
- Making & Crafting Experiments**
Hands-on crafting and mechanical model-building projects using simple materials, developing creativity, problem-solving skills and an early interest in transforming ideas into physical prototypes.

LEADERSHIP & ROLES

Patrol Member Scout Team, Daffodil International University	Member Robotics Club, Daffodil International University
Member Rotaract Club, Daffodil International University	Member CPC Club, Daffodil International University
Member Change Together Club, Daffodil International University	Member Daffodil Entrepreneurship Club, Daffodil International University

ACHIEVEMENTS & AWARDS

GOLD MEDAL Life Science Category Innovation World Cup (IWC), National Round Awarded for the Smart Helmet Safety System		
1st Place	Art & Drawing Bangladesh Atomic Energy Commission	2019
2nd Runner-Up	Script Writing Online School Competition	2021
Upazila-Level Award	Short Story & Poetry Writing	2016
8th Place	Robotics Project Daffodil International University RoboCamp	2025
Best Spoken Award	Nazrul's Division	2018